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Long-term recovery of epiphytic communities in the Great Bear Rainforest of coastal British Columbia

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ABSTRACT

The recent Great Bear Rainforest agreement recognises the high biodiversity values of this large intact area of coastal temperate rainforest by calling for old forest targets to be met by 2264. Recruiting young stands has joined conserving existing old stands as a strategy for achieving targets, but the point at which second growth stands recover oldgrowth attributes remains uncertain. We examined the recovery of epiphytes towards oldgrowth conditions by comparing community composition, richness and abundance between young (55–100 year old), mature (101–250 years old) and oldgrowth stands (>250 year old). We felled 77 western redcedar, amabilis fir, western hemlock and Sitka spruce trees, identified all epiphytes, and examined effects of stand age, region, tree species, site nutrient status and presence of residual trees on the epiphyte community. We found 229 taxa, including 49 bryophytes, 98 macrolichens and 82 crustose lichens. Epiphyte community varied by region and among tree species, but not by site productivity or presence of residual trees. In the northern region, trees in oldgrowth supported twice as many epiphyte species, seven times as many unique species, and a significantly different community composition for all functional groups (bryophytes, crustose lichens, hair lichens, cyanolichens and other macrolichens) relative to trees in stands younger than 200 years. Overall similarity between second growth and oldgrowth was about 50%. Young and mature stands overlapped considerably in richness, abundance, and community composition, indicating little recovery between 55 and 200 years. Our study suggests that in the northern region of the Great Bear Rainforest, epiphyte communities need more than 200 years to recover to oldgrowth conditions.

1. Introduction

One of the world's largest areas of relatively intact coastal temperate rainforest lies within British Columbia's (BC) 6.4 million hectare Great Bear Rainforest (DellaSala, 2011). This highly productive, biologically rich, and globally rare ecosystem is characterised by abundant precipitation and a moderate climate (DellaSala, 2011). Due to the extreme rarity of natural stand-replacing disturbances, particularly fire, temperate rainforests are dominated by ancient, structurally-complex forests that can be aged in millennia rather than centuries (Lertzman et al., 2002; Gavin et al., 2003; Daniels and Gray, 2006).

A recent agreement, endorsed by First Nations, the BC Government, environmental organisations and industry, aims to maintain ecological integrity in the Great Bear Rainforest by achieving high levels of oldgrowth ecosystem representation (generally 70% of each ecosystem over 250 years old) by the year 2264, and by limiting future forest harvest to 15% of the area (MFLNRO, 2016). Past logging, however, has already converted many productive ecosystems to even-aged second growth forests (e.g., productive western redcedar [*Thuja plicata*]-leading ecosystems have less than 10% oldgrowth; Holt and MacKinnon, 2006). Within the managed forest, plans call for 15% of each forest stand to be retained. Given the desire for sustainable, ecosystem-based management in the Great Bear Rainforest (Price et al., 2009; MFLNRO, 2016), important ecological questions include "How quickly will harvested stands develop oldgrowth characteristics?" and "How well will 15% in-stand retention maintain oldgrowth associates and facilitate recovery?"

Epiphyte communities are good indicators of ecological integrity, because of their slow recovery following disturbance and because of their various ecological functions as habitat, forage

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and nitrogen-fixers (Ellis, 2012). Wherever epiphytes have been studied in temperate and boreal ecosystems, old forests have supported a higher abundance and different communities of epiphytes than young or mature stands (e.g., Rose, 1976; Lesica et al., 1991; McCune, 1993; Neitlich, 1993; Goward, 1994; Esseen et al., 1996; Enns et al., 1999; Price and Hochachka, 2001; Campbell and Freeden, 2004). These changes have been related to potentially interacting variables including tree characteristics (e.g., substratum), stand characteristics (e.g., heterogeneity) and forest continuity (Ellis, 2012). Epiphyte communities in humid forests often undergo succession from early-colonising green foliose and crustose lichen species to hair lichens and finally cyanolichens and bryophytes (McCune, 1993; Sillett and Neitlich, 1996). Some cyanolichen species are sufficiently reliable oldgrowth associates that they have been used to indicate forest age and continuity (e.g., Goward and Pojar, 1998; Campbell and Fredeen, 2004).

Green tree retention (Rosenvald and Lõhmus, 2008) has been investigated as a strategy to maintain late successional epiphytes through forest harvest cycles by mitigating microclimate changes (Benson and Coxson, 2002) and dispersal limitations (Sillett and McCune, 1998). Results have varied. In coastal forests in Oregon, hair lichens and cyanolichens were more abundant in young stands with oldgrowth remnant trees (Peck and McCune, 1997), and the amount of oldgrowth-associate litter decreased with distance from remnants (Sillett and Goslin, 1999). However, in boreal ecosystems, remnant trees lost sensitive bryophytes and lichens in the first few years post-harvest likely due to seasonal drying (Perhans et al., 2009; Boudreault et al., 2013), and veteran trees supported a depauperate community of cyanolichens relative to their age-mates in adjacent old forest (Benson and Coxson, 2002; Radies and Coxson, 2004).

Here, we investigate the recovery of epiphyte communities in young and mature forests of the Great Bear Rainforest. Secondarily, we assess the potential effectiveness of remnant trees at facilitating recovery, but a retrospective study design, including sites with and without remnant trees, allows us to explore patterns rather than infer causation. We predicted that trees in oldgrowth would support unique epiphyte communities and that recovery would be slow. We predicted that oldgrowth-associated epiphytes would be positively affected by remnant trees, assuming that seasonal drying would pose a minor challenge in this maritime ecosystem. To understand epiphyte distribution among and within trees of the study area better, where there are known differences in tree composition and structure between second growth and oldgrowth stands (Banner and LePage, 2008; LePage and Banner, 2014), we also considered the effects of tree species and vertical canopy position on epiphyte communities.

This study forms part of an interdisciplinary investigation into the recovery trajectories of a variety of ecosystem attributes in coastal BC. As well as the epiphyte communities described in this paper, studies have examined stand structure and tree species composition (LePage and Banner, 2014), understory vegetation (Banner and LePage, 2008), forest growth and productivity, and soil properties including faunal communities.

2. Methods

2.1. Study area

The Great Bear Rainforest encompasses the coastal temperate rainforest of the central and northern BC coast (51°–55°N). Within this area, ecosystem recovery studies focussed on low elevation forests of the Very Wet Hypermaritime Coastal Western Hemlock biogeoclimatic subzone (CWHvh2; Banner et al., 1993) that grow on coastal islands and a mainland fringe between northern Vancouver Island and southeast Alaska. The mild, very wet (Banner et al., 1993) climate limits fire ignition and spread. Most stand-replacing natural disturbances are small and initiated by landslides, wind and alluvial flooding (Price and Daust, 2003; Banner et al., 2005). Gap-phase dynamics dominate stand regeneration, leading to uneven-aged oldgrowth stands that can be thousands of years old, considerably older than the oldest live tree (Lertzman et al., 1996; Daniels and Gray, 2006). Forests grow on moderate to steep slopes, while bogs cloak gentler terrain (Banner et al., 1993, 2005). Forested stands typically include a mix of western hemlock (*Tsuga heterophylla*), western redcedar, amabilis fir (*Abies amabilis*) and Sitka spruce (*Picea sitchensis*). Following stand-replacing disturbance, young CWHvh2 forests have a dense closed canopy for the first century. The canopy opens up between 100 and 160 years, and oldgrowth characteristics (large old trees, downed wood, snags, vertical and horizontal heterogeneity) develop over the next hundred years (Franklin et al., 2002; LePage and Banner, 2014).

Small-scale commercial harvesting in the Great Bear Rainforest began about a century ago; most cutblocks harvested were small (<20 ha; many much smaller); today, they contain residual trees (small and less desirable species left after logging), and are surrounded by oldgrowth. Although stands vary in their development, 80–100 year-old harvested stands in the study area have a 40–55% structural similarity to oldgrowth stands, with nutrient-rich sites developing large trees faster than mesic sites (LePage and Banner, 2014).

2.2. Sample sites

The sites were located in two accessible areas, one in the north and one in the central region of the Great Bear Rainforest (Fig. 1). Second growth site selection was severely limited due to the paucity of stands meeting age criteria (young: 55–100 years old; mature: 101–250 years old) in the region. Second growth stands (>1 ha of uniform stand characteristics), accessed from the ocean and narrow inlets, were identified using maps, local knowledge, and visual inspection. Oldgrowth stands (>250 years old) in the northern region were accessed from roads near the ocean and were selected to match plant community characteristics of second growth stands. As a group they were slightly farther from the ocean and higher in elevation (2.2–2.4 km and 33–189 m elevation) than the second growth sites (0.1–0.5 km and 1–59 m elevation) due to timing and access logistical limitations. All sites were sheltered by islands rather than exposed to the open Pacific, controlling somewhat for oceanic influence. Slope and aspect varied (0–86% and 12–354°, respectively). The 29 sampled sites included 11 young stands initiated by harvesting, 13 mature stands initiated by harvesting, windthrow or fire (sometimes with a mix of disturbance types) and five uneven-aged oldgrowth stands.

Northern sites were colder than central sites (mean annual temperature: 6.9 ± 0.06 °C and 8.1 ± 0.02 °C respectively), and had slightly higher mean annual precipitation (3200 ± 105 and 3000 ± 40 mm/year respectively), with relatively less spring precipitation and more summer and fall precipitation (interpolated 1960–1990 climate normals for sites were extracted from ClimateBC; Wang et al., 2016).

At each site, we cored four of the largest diameter, non-residual trees and defined stand age as the age of the oldest cored tree. Oldgrowth stands were assigned an age of 250+ years because rotten tree cores prevented accurate tree ring counts and because stands are frequently older than the oldest tree. We determined site productivity, based on the BC biogeoclimatic ecosystem classification system (Banner et al., 1993; Kranabetter et al., 2003). Most sites had sub-mesic to mesic soil moisture (two were hygric), but varied in nutrient regime. We grouped sites (using Banner et al., 1993) into "rich" (CWHvh2/05 and /06 site series), "medium"

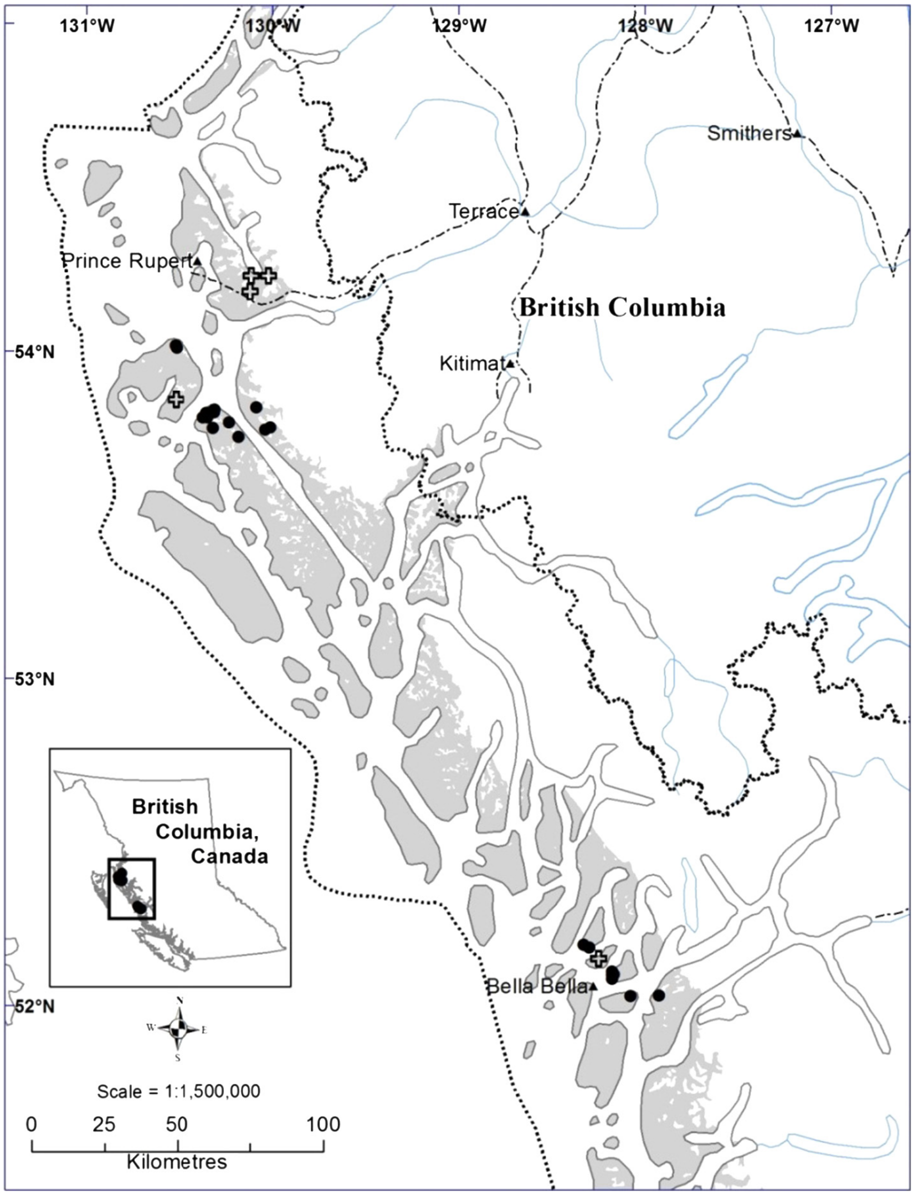

Fig. 1. Study area (circles show second growth and crosses show oldgrowth sites) within the Great Bear Rainforest (dotted outline). The Coastal Western Hemlock zone, Very Wet Hypermaritime subzone, central variant (CWHvh2) is shaded.

(CWHvh2/04 site series) and "poor" (CWHvh2/01 site series) productivity classes.

Presence of residual trees was based on density estimates completed for stand structure studies in the second growth stands (two cruise plots per site; methods described in LePage and Banner, 2014); oldgrowth stands include residual trees by definition. Epiphyte load was estimated from the ground on all trees in the two plots (five classes: absent, rare, occasional, common, abundant) to test the assumption that residual trees would harbour higher abundance. About one-third of young stands (4/11) and two-thirds of mature stands (9/13) had large residual trees (co-dominant canopy trees >70 cm dbh) and 18% of young (2/11) and 31% (4/13) of mature stands had even larger veteran trees (dominant trees that stand above the mean canopy).

Complete epiphytic communities were described from 2 to 4 trees felled in each site (total 77 trees). Sample trees were selected as healthy co-dominant trees representing the species present in the plot that could be felled safely. Few sites supported all four tree species: nutrient-poor sites had more western redcedar and less amabilis fir; rich sites had more Sitka spruce. Western hemlock was present in all sites but one. Trees in the northern region were larger than those in the central region (Table 1). Further information on structure in the second growth sites is available in LePage and Banner, 2014.

The sites in the northern region were sampled in 2004–2008. The central region was sampled in 2006, and the sample was highly unbalanced, due to reasons beyond our control (Table 1). Because this sample includes only a single oldgrowth site, we focus analyses of recovery on the northern sample, and hence limit most conclusions to this region. Mature sites in the northern region were all <200 years old, and so we limit discussion of recovery to 200 years.

2.3. Epiphyte sampling

We recorded the presence and collected samples of each epiphyte species from the trunk, large branches and small twigs of each felled tree (3–4 h effort per tree). We divided the tree into

Table 1
Sample size, stand basal area and sample tree size (mean ± SD) in the northern and central regions of the Great Bear Rainforest (Young = 55–100 years old; Mature = 101–250 years old; Oldgrowth > 250 years old).

	Central region			Northern region			Total
	Young	Mature	Oldgrowth	Young	Mature	Oldgrowth	
No. of sites	2	7	1	9	6	4	29
No. of trees							
Amabilis fir	–	2	–	5	3	5	15
Sitka spruce	2	1	–	9	4	4	20
Western hemlock	2	5	1	8	7	4	27
Western redcedar	–	8	1	2	1	3	15
Stand basal area (m^2)	61 ± 3	84 ± 16	n/a	73 ± 20	74 ± 15	n/a	
Tree height (m)	30 ± 7	24 ± 4	23 ± 1	32 ± 6	34 ± 3	35 ± 5	
Tree diameter (cm)	44 ± 14	46 ± 12	n/a	53 ± 13	66 ± 17	n/a	

lower, middle and upper segments (equal sized thirds; mean segment length = 10.1 m; range = 7–15 m) for a total of 231 segments. All unfamiliar samples were submitted to BC experts for identification. Taxonomy and nomenclature follows Esslinger (2016) for lichens and Flora of North America Editorial Committee (2007, 2014) and Frey et al. (2006) for bryophytes. Voucher specimens are housed at the University of BC herbarium, Vancouver. We also estimated abundance by functional group (bryophytes, hair lichens, cyanolichens, and other lichens including crustose, green-algal foliose and fruticose lichens) in each segment of the tree: 0 = absent; 1 = rare, 1–3 thalli; 2 = occasional, 4–10 thalli; 3 = common, >10 thalli; and 4 = abundant, > 50% of branches.

2.4. Data analysis

We compared community composition, species richness, and abundance of epiphytes across three age classes (young, mature, oldgrowth) and four tree species (amabilis fir, Sitka spruce, western hemlock, western redcedar). We also examined effects of site productivity (rich, medium, poor) and presence of residual and veteran trees on epiphytes. Most analyses combined vertical segments and used the whole tree as the treatment unit. For species richness and community composition analyses, we grouped species into functional groups: bryophytes, crustose lichens, hair lichens, cyanolichens and other macrolichens (including green-algal foliose and fruticose lichens).

2.5. Community composition

We used PRIMER-E software (v6; Clarke and Gorley, 2006) for multivariate community analyses. We calculated a community resemblance matrix using the Sorensen index and used non-metric multidimensional scaling (NMDS) to represent similarities visually. We then used Sorensen index to compare overall similarity among age classes (Faith et al., 1987). We tested for community differences associated with environmental variables using the ANOSIM routine, a permutation and randomisation test based on ranks within the full dimensional similarity matrix. The test statistic, R varies from 0 to 1, with values close to one indicating good separation of communities and values near zero showing close similarity. Significance testing compares the observed R to its permuted distribution (based on maximum possible or 999 permutations). Where global tests indicated ecologically meaningful and statistically significant differences, we examined R-statistics for pairwise comparisons (with a Bonferroni correction).

2.6. Richness and abundance

To understand the effects of age class, tree species, site productivity, and the presence of residual and/or veteran trees on species richness and abundance in whole trees, we developed mixed-effects models with the 'nlme' package in R version 3.0 (R Development Core Team, 2015; Pinheiro et al., 2015). We treated the data as a completely randomized split plot design where the main plot factor was age class and the split plot factor was tree species (with the random effect site nested in ageclass acting as the error term for the main-plot factor).

$$Y_{ijk} = \mu + \text{AgeClass}_i + \text{Site}_{(i)j} + \text{TreeSpecies}_k + (\text{TreeSpecies x AgeClass})_{ik} + \varepsilon_{(i)jk}$$

Residuals were normally distributed but had heterogeneous variances among age classes, which we incorporated in the model. The model was extended to include the observational covariate of site productivity and possible influence of residual and veteran trees (the latter two for the mature and young age class data only), but when these factors had no significant effect on epiphyte richness or abundance we retained the simpler model.

To examine epiphyte richness and abundance across the vertical tree segments, we used a completely randomized split-split plot mixed-effects model. We ran whole-tree and vertical-segment models on total species richness, richness of five functional groups and abundance of four groups. We wrote custom orthogonal contrasts using the 'multcomp' package in R (Hothorn et al., 2008).

2.7. Indicator species

To explore which species consistently contributed most to dissimilarities among communities, we used the SIMPER routine in PRIMER-E (Clarke and Gorley, 2006); for species associated with oldgrowth trees, we compared presence in each age class with Fisher Exact Tests (with Monte Carlo simulation). For the subset of oldgrowth associates also present on second growth trees, we tested for an effect of residual trees.

3. Results

3.1. Community composition

Sampled trees supported 229 epiphyte taxa, including 49 bryophytes, 82 crustose lichens, 14 hair lichens, 39 cyanolichens and 45 other macrolichens (Supplementary Materials Table S1). Of these, nearly half (109 species) were collected one to three times. Across the whole region, we found 68 species only on oldgrowth, 12 only on mature and 14 only on young trees. The richest tree, an oldgrowth amabilis fir, hosted 71 species.

Northern and central regions supported different epiphytic communities (Fig. 2; region effect: R = 0.74, $P = 0.001$; age effect: Global R = 0.37, $P = 0.001$; ANOSIM by region and tree age). Eighty-nine species were found only in the northern region

(N = 55 trees) and 34 species only in the central region (N = 22 trees). Northern samples included more than twice as many cyanolichen species (36 vs. 14), and more bryophytes, crustose, hair and other macrolichen species.

Within the northern region, epiphyte communities varied non-linearly with tree age class (Global R = 0.47, $P < 0.001$; ANOSIM by tree age and species). Oldgrowth communities separated from young and mature communities, while the latter two overlapped (Fig. 2), northern region only; oldgrowth versus mature: R = 0.73, $P < 0.001$; oldgrowth versus young: R = 0.80, $P < 0.001$; mature versus young: R = 0.08, $P = 0.2$). This pattern held for all functional groups. Crustose lichens and cyanolichens showed least overall similarity between second growth and oldgrowth; bryophytes and other macrolichens showed the most (Table 2).

Western redcedar hosted different epiphyte communities than other tree species, which all hosted similar communities (Global R = 0.23, $P < 0.001$; pairwise tests including western redcedar had R-values ≥ 0.57; R-values for other tests were all < 0.16). Some functional groups were more closely related to tree species than others. Neither hair nor crustose lichens varied with tree species. Bryophytes and other macrolichens differed between western redcedar and other species (pairwise tests including western redcedar had R-values between 0.38 and 0.7; R-values for other tests were all < 0.09). Oldgrowth amabilis fir supported a distinct cyanolichen community (pairwise R-values 0.37–0.73). Amabilis fir was particularly rich in unique species, hosting at least three times as many as other trees, including 17 unique cyanolichens, all but two limited to oldgrowth (Supplementary Material Figure S1).

Although residual trees supported higher epiphyte loads (Fig. 3), there was no indication that these residual trees (measured at the stand-level) affected epiphyte community composition in regenerating trees (effect of large residuals: R = 0.009, $P = 0.4$; effect of veteran trees: R = 0.018, $P = 0.4$; one-way ANOSIM). Neither did epiphytic community vary with site productivity (R = −0.05, $P = 0.7$; ANOSIM by age class and site productivity). Although western redcedar was more frequent, and amabilis fir less frequent, on poor sites (and both hosted different epiphyte communities for some functional groups), there was no additional effect of site productivity on epiphyte communities (R = 0.04, $P = 0.3$ for ANOSIM by tree species and site productivity).

Table 2
Sorensen's percent similarity comparisons in the northern region based on presence-absence of species within each age class (Young: 55–100 years; Mature: 101–200 years; Oldgrowth: >250 years; N = 24, 15 and 16 respectively).

Functional group	Young vs mature	Young vs oldgrowth	Mature vs oldgrowth
All	82	51	48
Bryophytes	82	67	60
Crustose lichens	81	32	34
Cyanolichens	74	29	26
Hair lichens	57	36	50
Other macrolichens	87	74	68

The best multivariate explanatory model describing community patterns across all northern sites included only tree age (ρ = 0.45, P < 0.001; BEST). The best explanatory model describing patterns across young and mature sites included only tree species (ρ = 0.26, p < 0.001; BEST).

3.2. Epiphyte richness and abundance

Oldgrowth trees supported about twice as many epiphyte species as young or mature trees in second growth stands (Fig. 4; $F_{2,25} = 23.2$, $P < 0.001$). All functional groups were richer on oldgrowth trees, with cyanolichens and crustose lichens increasing more than other macrolichens and bryophytes (Fig. 4a). Crustose lichens were the only group that showed any tendency to increase between young to mature trees ($P = 0.13$; orthogonal contrasts). There was no significant effect of tree species on richness (Fig. 4b; $F_{3,25} = 1.5$, $P = 0.25$) and no tree species and age class interactions ($F_{6,25} = 0.4$, $P = 0.9$).

Richness of two functional groups varied among tree species (Table 3): amabilis fir hosted more than twice as many cyanolichen species and western redcedar hosted about half as many other macrolichens as other tree species. Cyanolichen richness was particularly high on oldgrowth amabilis fir which hosted six to nine more species per tree than trees of other age class or species.

Abundance of bryophytes, hair lichens and cyanolichens was highest in oldgrowth trees, while the combined abundance of

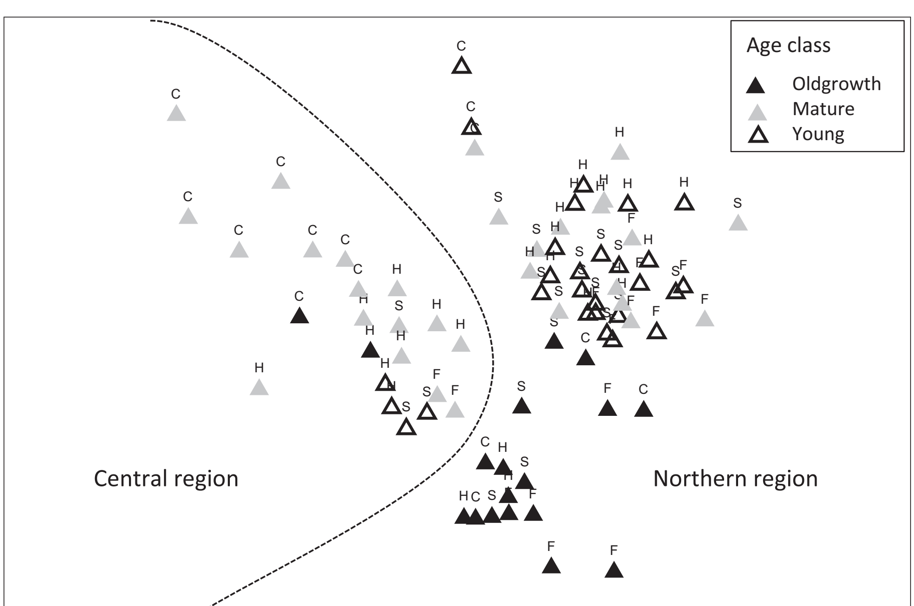

Fig. 2. NMDS plot of epiphyte community similarity on whole trees (N = 77) showing the effect of tree age class, region and tree species in the Great Bear Rainforest. Labels show tree species (F = amabilis fir; C = western redcedar; H = western hemlock; S = Sitka spruce).

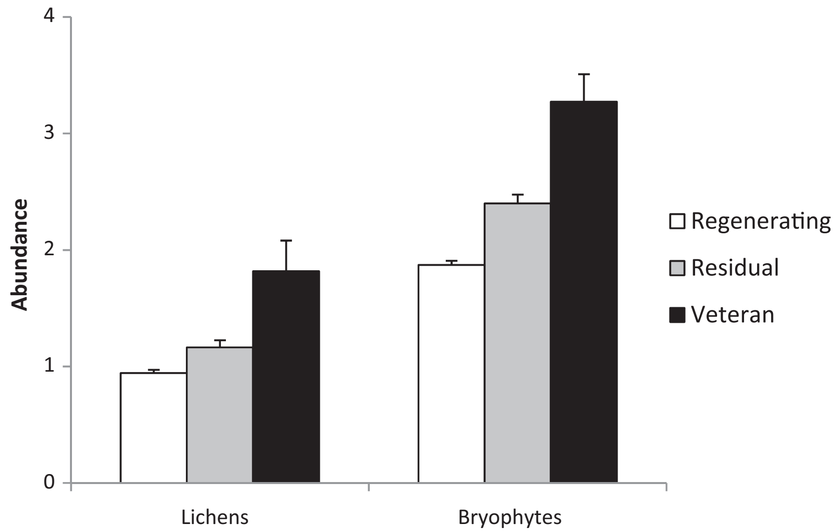

Fig. 3. Abundance of epiphytic lichens and bryophytes (mean + SE) on regenerating co-dominants, residual co-dominants and large veteran trees in cruise plots (N = 344 regenerating, 110 residual and 11 veteran trees). Tree types differed significantly (all comparisons $P < 0.001$ Tukey's hsd; ANOVA with site as random effect).

crustose and other macrolichens was high for all age classes (Table 4; Fig. 5). Abundance varied with tree species for some functional groups mirroring richness patterns: amabilis fir had the highest abundance of cyanolichens, western redcedar had a lower abundance of crustose and other macrolichens, and western hemlock had slightly lower bryophyte abundance (all $P < 0.01$, orthogonal contrasts). There was a significant interaction between age class and tree species for cyanolichens: most oldgrowth amabilis fir trees had more than 10 thalli, while younger fir and other tree species hosted 3 or fewer thalli (Table 4 and Fig. 5).

Neither site productivity nor the presence of residual trees improved the model for richness or abundance of any functional group.

3.3. Indicator species

In the northern region, 100 epiphyte species were found only on oldgrowth trees (with ten species only on young and four only on mature trees); most of these species (63%) were found on fewer than four trees, and hence less useful indicators. Of the sixteen species contributing most to the dissimilarity between oldgrowth and second growth trees, and found on more than half the trees of at least one age class, all but one species (*Hypogymnia apinnata*) were more common in oldgrowth trees, and six were limited to oldgrowth (Table 5). Of the 83 species accounting, cumulatively, for 50% of the dissimilarity between oldgrowth and second growth communities, over three quarters (64 species) were most common on oldgrowth trees.

The subset of nine species found most often on, but not limited to, oldgrowth trees (bold in Table 5) and therefore potentially facilitated by residual trees in second growth stands were equivocally associated with residual trees. As a group, the community of these species differed slightly between trees in second growth stands with and without residuals ($R = 0.13$, $P = 0.01$; ANOSIM). This difference was driven by the higher frequency of *Lobaria oregana, Cladonia ochrochlora*, *Hypogymnia duplicata* and *Platismatia lacunosa* on young and mature trees in stands with residuals; no crustose lichens or bryophytes were more common in stands with residuals. Individually, however, no species was present significantly more often on trees in stands with than without residuals (*Alectoria sarmentosa* $P = 0.11$, *H. duplicata* $P = 0.11$, *L. oregana* $P = 0.16$; all other P-values > 0.3; Fisher's exact test with Monte Carlo simulation).

3.4. Vertical distribution

Epiphyte communities varied vertically in a similar pattern in all age classes (Global $R = 0.48$, $P = 0.001$; ANOSIM by segment and age class; Fig. 6). Communities on the bottom third of trees were most different; middle communities were intermediate between upper and lower (low versus middle: $R = 0.44$, low versus upper: $R = 0.76$, middle versus upper: $R = 0.23$; all $P = 0.001$).

Bryophyte species richness declined considerably with height, while crustose, hair, and other macrolichen species richness increased with height, and cyanolichens peaked in the middle segment (Fig. 7a, Supplementary Materials Table S2). Patterns were similar across age classes. Changes in abundance with height for hair, crustose and other macrolichens matched species richness patterns (Supplementary Materials Table S3). The strong decrease in bryophyte richness with height was not mirrored by abundance (Fig. 7b), showing that the upper canopy bryophyte community was dominated by a few species, including *Antitrichia curtipendula*, *Douinia ovata*, *Frullania nisquallensis* and *Iwatsukiella leucotricha*.

4. Discussion

4.1. Effect of stand age

In the hypermaritime Coastal Western Hemlock ecosystem of the northern region of the Great Bear Rainforest, co-dominant trees in oldgrowth stands supported different epiphyte communities than those in second growth stands from 55 to 200 years old. Oldgrowth communities had twice as many species, with higher species richness of every functional group, and a higher abundance of bryophytes, cyanolichens and hair lichens. Conversely, communities in young and mature stands overlapped considerably, suggesting that, in these ecosystems, epiphyte communities develop slowly, with little recovery between 55 and 200 years. While studies in a variety of ecosystems have found that epiphyte assemblages differ and abundance increases with stand age (e.g., McCune, 1993; Neitlich, 1993; Esseen et al., 1996; Goward and Pojar, 1998; Pipp et al., 2001; Price and Hochachka, 2001; Berryman and McCune, 2006; Bartels and Chen, 2015), most have found that mature stands host communities that are intermediate between those in young and in oldgrowth stands. Total species richness continues to increase with age in some ecosystems (e.g., Campbell and Fredeen, 2004), but, more commonly, asymptotes

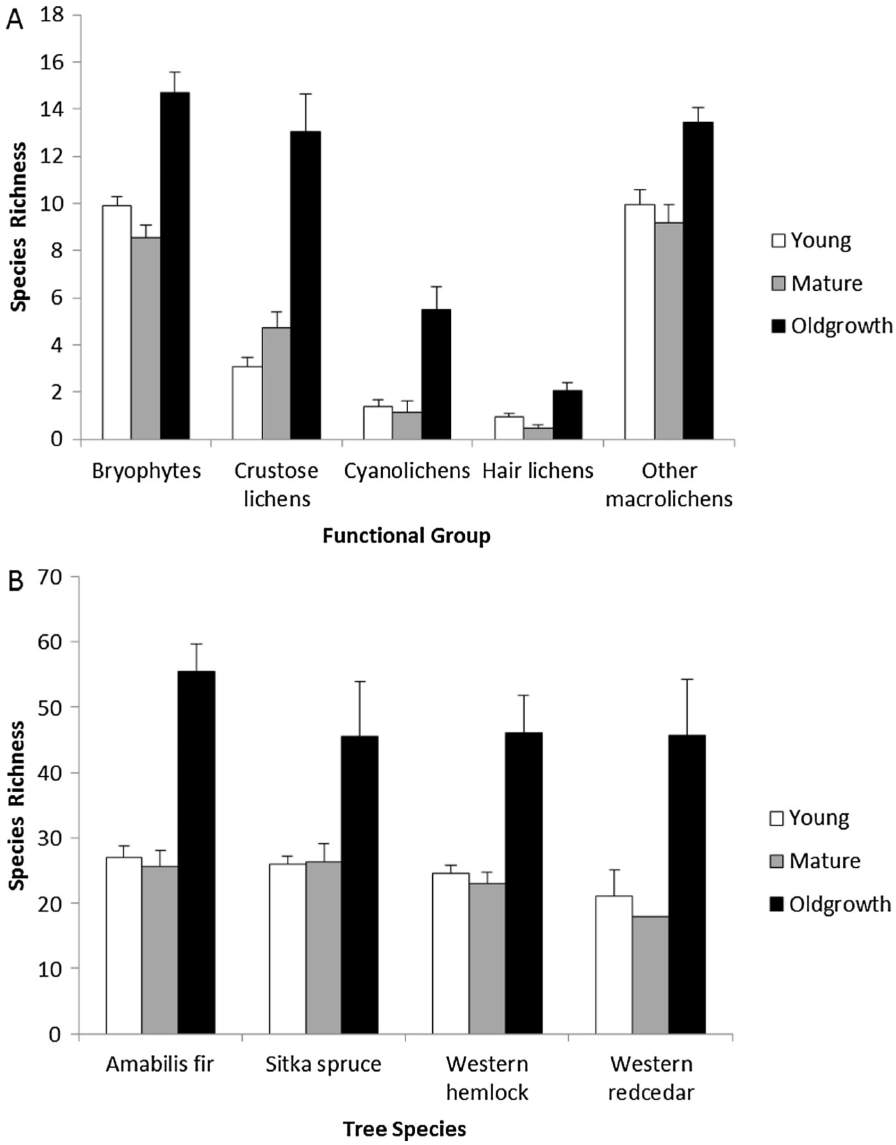

Fig. 4. Epiphyte species richness (mean + SE) by functional group (A) and by host tree species (B) in each age class (Young: N = 24; Mature: N = 15; Oldgrowth: N = 16; see Table 1 for N by tree species and age class).

Table 3
Significance values for the effects of tree age class, tree species and their interaction on the richness of epiphyte functional groups. Bold values show $P < 0.05$.

Group	Tree age class		Tree species		Interaction	
	$F_{2,25}$	P	$F_{3,25}$	P	$F_{6,25}$	P
Bryophytes	15.41	**<0.001**	1.24	0.3	0.66	0.7
Crustose lichens	14.19	**<0.001**	0.78	0.5	1.37	0.3
Cyanolichens	9.60	**<0.001**	9.00	**<0.001**	3.56	**0.01**
Hair lichens	8.85	**0.001**	1.76	0.18	2.36	0.06
Other macrolichens	7.89	**0.002**	3.31	**0.04**	1.09	0.4

Table 4
Significance values for the effects of tree age class, tree species and their interaction on the abundance of epiphyte functional groups in the northern region of the Great Bear Rainforest. Bold values show $P < 0.05$.

Group	Tree age class		Tree species		Interaction	
	$F_{2,25}$	P	$F_{3,25}$	P	$F_{6,25}$	P
Bryophytes	4.45	**0.02**	4.47	**0.01**	0.78	0.6
Cyanolichens	3.81	**0.04**	7.75	**<0.001**	3.11	**0.02**
Hair lichens	4.10	**0.02**	1.68	0.2	1.89	0.12
Other lichens	0.05	1.0	13.34	**<0.001**	1.79	0.14

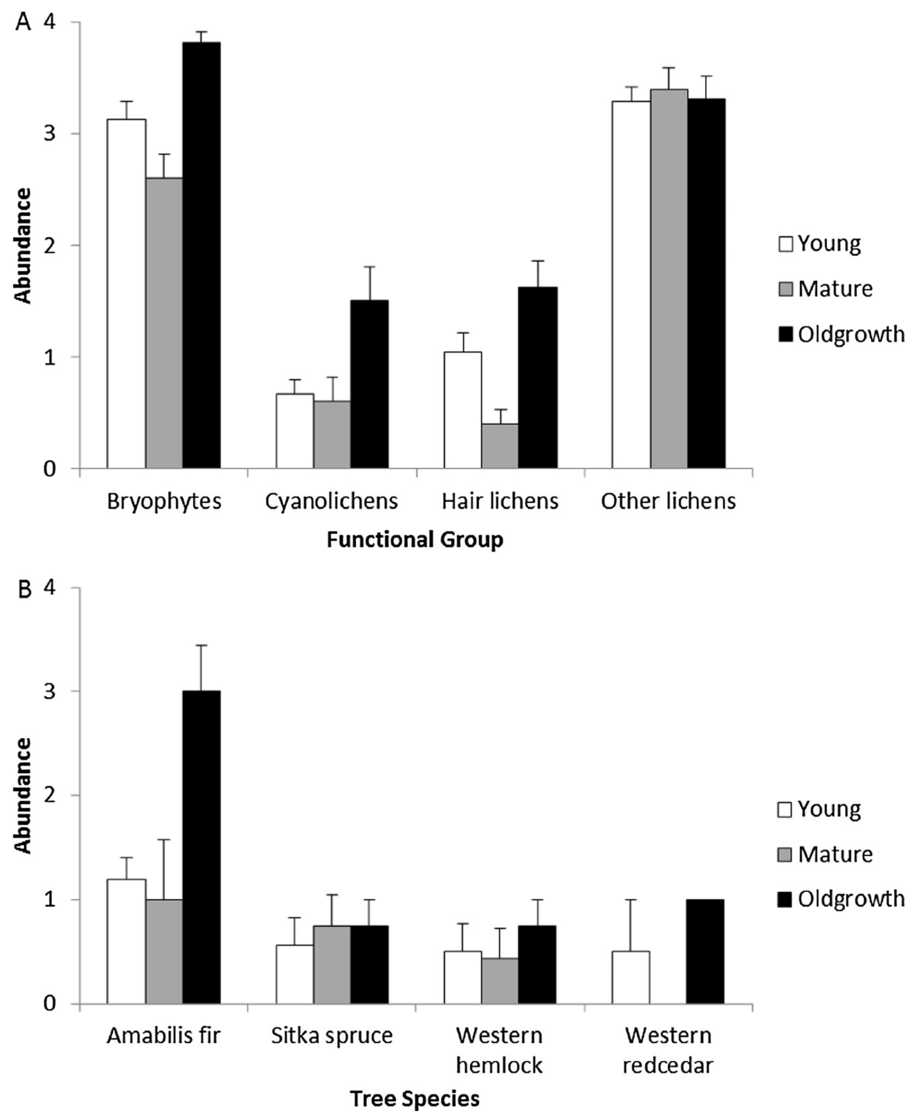

Fig. 5. Epiphyte abundance (mean + SE) in functional groups (A) and cyanolichen abundance on different tree species (B) by age class (Young: N = 24; Mature: N = 15; Oldgrowth: N = 16; see Table 1 for N by tree species and age). Abundance classes: 1 = 1–3 thalli; 2 = 4–10 thalli; 3 = >10 thalli; 4 = present on >50% of branches (rating for vertical segment with highest abundance).

Table 5
Species contributing most to the dissimilarity between epiphyte communities growing on young, mature, and oldgrowth trees (N = 24, 15 and 16 respectively) in the northern region of the Great Bear Rainforest. All species listed had a mean dissimilarity $\geq$ 0.7 and dissimilarity/SD $\geq$ 1; all were significantly differently distributed (P < 0.02 Fisher's Exact Test with Monte Carlo simulation for P-value). Bold species were used for analysis of residual trees.

Group	Species	Mean abundance		
		Young (55–100 years)	Mature (101–200 years)	Oldgrowth (>250 years)
Bryophytes	***Bazzania denudata***	0.08	0.13	0.81
	Herbertus aduncus	0.04	0	0.69
	Iwatsukiella leucotricha	0	0	0.63
Crustose lichens	***Lopadium disciforme***	0.04	0	0.69
	Micarea prasina s. lat.	0	0	0.56
	Micarea prasinella	0	0	0.56
	Mycoblastus sanguinarius	0.08	0.33	0.69
	Thelotrema lepadinum	0	0	0.63
Cyanolichens	*Leptogidium dendriscum*	0	0	0.69
	Lobaria oregana	0.29	0.27	0.69
Hair lichens	***Alectoria sarmentosa***	0.75	0.33	0.94
Other macrolichens	***Cladonia ochrochlora***	0.08	0	0.88
	Hypogymnia apinnata	0.92	0.87	0.38
	Hypogymnia duplicata	0.08	0.13	0.75
	Platismatia lacunosa	0.17	0	0.69
	Sphaerophorus venerabilis	0	0	0.88

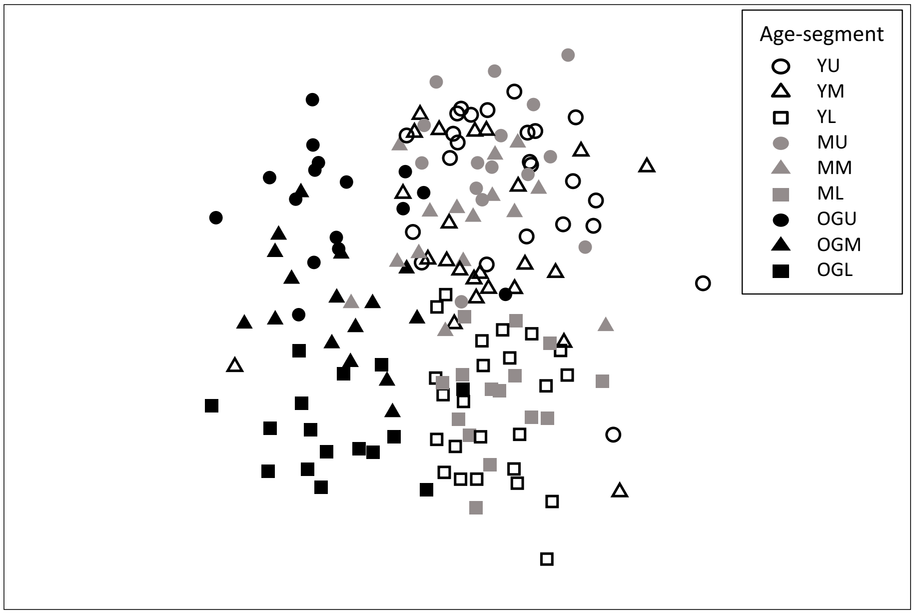

Fig. 6. NMDS plot of epiphytic community similarity by vertical tree segment and age class in the northern region of the Great Bear Rainforest (N = 165 segments). Shading represents tree age class (Y = 55–100 years; M = 101–200 years; OG > 250 years); shape represents segment (L = lower; M = middle; U = upper third of tree). Figure rotated so that vertical segment aligns with Y-axis.

earlier (e.g., Dettki and Esseen, 1998; Goward and Pojar, 1998; Peterson and McCune, 2001; Boudreault et al., 2002; Bartels and Chen, 2015), so that oldgrowth communities are not necessarily more diverse. Our results are unusual in finding that epiphyte communities on mature trees (aged 101–200 years) are not intermediate in community composition, abundance or richness between young and oldgrowth trees.

Changes in epiphyte communities with stand age have been related to a suite of interacting tree-level and stand-level variables (Ellis, 2012). Essentially, as stands age, increased structural heterogeneity provides greater habitat diversity, and time provides opportunity for epiphytes to disperse and establish. In the rarely disturbed rainforests of BC, stand continuity is extremely long (Gavin et al., 2003; Daniels and Gray, 2006), allowing complex structure to develop over millennia, and giving ample time for community development. Perhaps long, slow stand-development trajectories account for our results.

In parallel studies, Banner and LePage (2008) and LePage and Banner (2014) found that understory plant communities and stand structure recovered faster than we found for epiphyte communities. We suggest two hypotheses for the discrepancy. Methodologically, their structural indicator (trees > 75 cm lumped into one class) does not capture the very large trees and gaps that are related to increased habitat heterogeneity and epiphyte diversity elsewhere (Neitlich and McCune, 1997), and that continue to develop in these ecosystems. Ecologically, stand structure recovery does not immediately translate into colonisation; poor dispersal of some species means that communities develop slowly, such that some species (including *Lobaria oregana*, one of the oldgrowth associates in our study) can indicate antique forests (Goward and Pojar, 1998). The more rapid recovery of stand structure and understory vegetation on nutrient-rich sites (Banner and LePage, 2008; LePage and Banner, 2014) was not mirrored by epiphyte communities, perhaps due to similar reasons.

We are reasonably confident that the pattern we describe is due to stand age, and associated variables, rather than to unrelated site attributes. We were able to control some attributes through selection of site locations: all sites (within the northern region) had submesic to mesic moisture regime within the same biogeoclimatic subzone; no sites had co-dominant deciduous species (avoiding potential drip-zone effects; Goward and Arsenault, 2000; Campbell et al., 2010); all age classes had similar canopy height. Although oldgrowth sites were slightly higher in elevation and somewhat further from the ocean, all sites were low elevation and near, yet not directly exposed to, the ocean. Sites varied in nutrient regime and tree species, but representation in the northern region was similar between oldgrowth and second growth sites. Unbalanced attributes precluded reliable analyses of age effects for the central region.

Although we found slow recovery in all second growth sites in the northern region, we found two young sites with exceptionally diverse communities, including many cyanolichens, in the central region. Unlike other sites in either region, these ecosystems had hygric soil moisture and were located on flat slopes directly adjacent to the ocean; attributes related to increased diversity, particularly of cyanolichens, elsewhere (Price and Hochachka, 2001; Peterson and McCune, 2003; Radies et al., 2009). One site had severely fluted and corkscrewed trees, a formation associated with coastal toe slopes and old village sites, and a possible indicator of calcium enrichment (Julin et al., 1993, J. Pojar personal communication). Although climatic differences likely account for some community composition differences between regions, these two sites suggest that species richness can also be high in the central region and that climate is not necessarily a limiting factor.

Although oldgrowth trees supported more species of all functional groups, the relative increase varied, with cyanolichens differing most with age, consistent with other studies (McCune, 1993; Sillett and Neitlich, 1996; Benson and Coxson, 2002; Campbell and Fredeen, 2004). Cyanolichens depend on particular microclimatic conditions with sufficient moisture, including rain, moderate ventilation and filtered light (Stevenson et al., 2011; Gauslaa, 2014). Cyanolichens did not colonise regenerating trees in small gaps in BC interior rainforest, indicating microclimate or substrate limitations (Benson and Coxson, 2002; Radies and Coxson, 2004). Conversely, transplanted cyanolichens grew in second growth ecosystems in Oregon, indicating dispersal limitations (Sillett and McCune, 1998; Sillett et al., 2000). Campbell and Fredeen (2004) suggest that dispersal limits colonisation of young

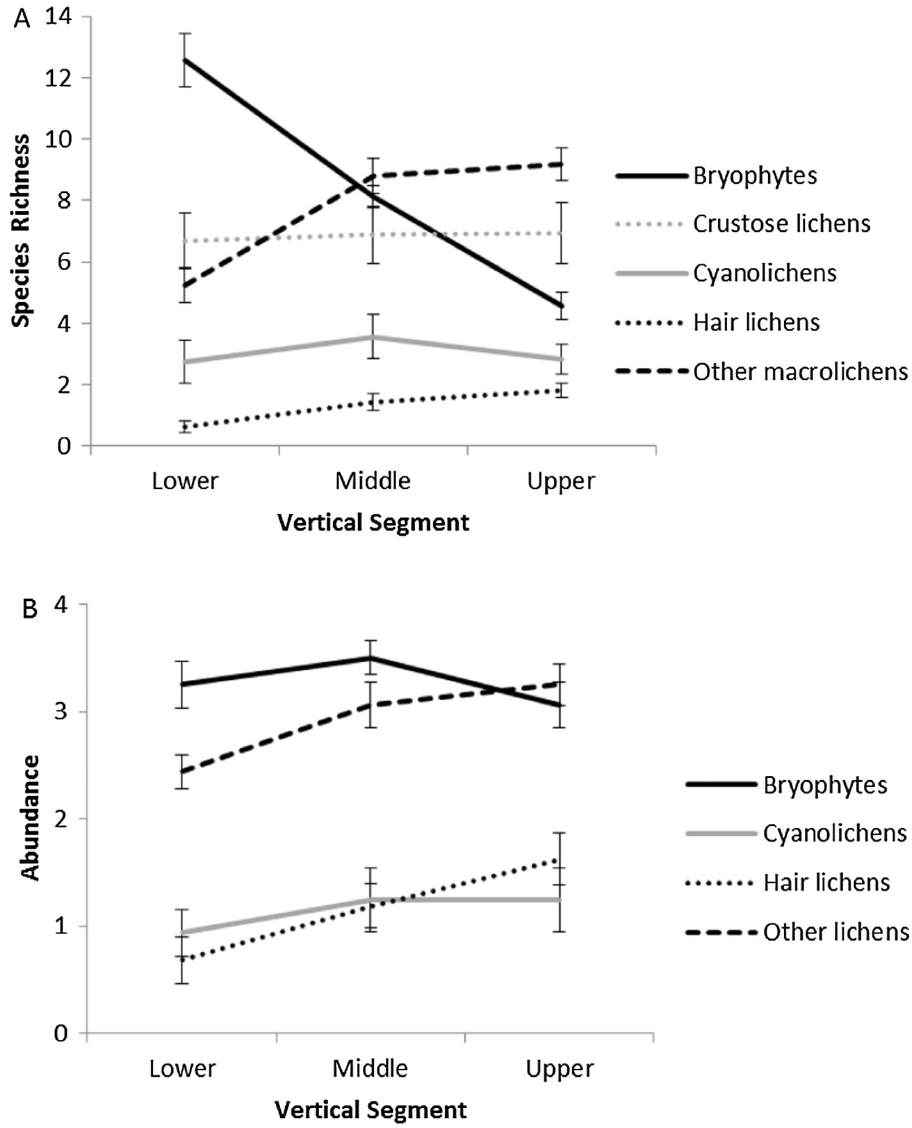

Fig. 7. Richness (A) and abundance (B) of each epiphyte functional group (mean ± SE) in the lower, middle and upper third of oldgrowth trees (N = 16). Abundance classes: 1 = 1–3 thalli; 2 = 4–10 thalli; 3 = >10 thalli; 4 = present on >50% of branches.

stands and that microclimate limits cyanolichen diversity in mature stands.

Unsurprisingly, we found species associated with oldgrowth; however, we also found species in younger forests that are limited to oldgrowth elsewhere. *Lobaria oregana* is a well-established oldgrowth associate in wet temperate ecosystems (McCune, 1993; Sillett and Neitlich, 1996; Goward and Pojar, 1998; Peterson and McCune, 2001), and showed the same pattern in our sites. Similarly, *Alectoria sarmentosa* is recognised as an oldgrowth associate in the US Pacific Northwest and in Europe (McCune, 1993; Esseen et al., 1996; Neitlich and McCune, 1997; Peck and McCune, 1997; Peterson and McCune, 2001); it was more common in our old sites, but occurred also in young and mature sites. Some species associated with old forest in inland rainforests (e.g., *Cavernularia hultenii*, *Sphaerophorus tuckermanii* [previously *S. globosus*], *Hypogymnia enteromorpha*, *Platismatia norvegica*; Goward and Pojar, 1998) were common in all ages in our hypermaritime stands, while *Leptogidium dendriscum* (formerly *Polychidium*) remained limited to old forest. *Usnea longissima*, considered to be dispersal-limited (Esseen et al., 1981; Keon and Muir, 2002), was found in young, mature and oldgrowth stands, including one 70 year old site with no residual trees, suggesting that dispersal from nearby oldgrowth can occur within 70 years. In our study, *Hypogymnia duplicata* was associated with oldgrowth while *H. appinata*, associated with old forest in inland BC rainforests, was significantly associated with second growth. In Alaska *H. duplicata* is distributed in oceanic and *H. appinata* in suboceanic conditions (Root et al., 2014).

4.2. Effect of tree species

We found a smaller effect of tree species on epiphyte communities: old amabilis fir trees hosted the most cyanolichens, and western redcedar hosted a different community with fewer other macrolichens. An effect of tree species is well established in other ecosystems, although particular associations may interact with larger-scale factors including climate and topography to produce complex patterns that are difficult to transfer. For example, in southern BC, high-elevation amabilis fir stands supported high cyanolichen biomass while low-elevation hemlock coastal stands were dominated by hair lichens (Price and Hochachka, 2001), whereas in Oregon, high-elevation amabilis fir stands supported a low cyanolichen biomass and higher hair lichen biomass than the low-elevation hemlock stands (Berryman and McCune, 2006).

Cyanolichens' narrow moisture and light tolerances may overwhelm substrate preferences.

4.3. Vertical gradient

Epiphyte communities changed along a vertical gradient, as described for boreal, temperate and tropical ecosystems worldwide (Ellis, 2012). Bryophytes dominated the lower segment of each tree, consistent with the "similar gradient hypothesis" (McCune, 1993; McCune et al., 1997). However, in these very wet ecosystems, bryophytes were common throughout each tree similar to the oldest and wettest forests of the Pacific Northwest US (Sillett and Neitlich, 1996). The distribution of epiphytic moss mats in these coastal rainforests determines habitat availability for other species, including nesting platforms for marbled murrelets (Burger et al., 2010). Hair lichens increased towards the top of each tree, but only achieved high abundance in a few scattered trees, whereas, further south, *Alectoria sarmentosa*, the most abundant hair lichen in our samples, contributes considerable biomass in the mid to upper canopy (McCune et al., 2000). *Bryoria*, abundant in drier forests, were rare in these rainforests, likely due to their sensitivity to extended hydration (Coxson and Coyle, 2003). Cyanolichens reached their maximum richness in the middle segment of trees, but, in common with other low elevation coastal forests in the Pacific Northwest (Neitlich and McCune, 1997; Price and Hochachka, 2001), had low abundance and richness throughout.

4.4. Effect of residual trees

Green tree retention can be an effective strategy to maintain within-stand biodiversity in harvested forests (Rosenvald and Lõhmus, 2008). Residual trees support some epiphytic species (sometimes not cyanolichens; Benson and Coxson, 2002) through disturbance as well as provide increased heterogeneity that promotes community development on regenerating trees (Sillett and Goslin, 1999; Coxson and Stevenson, 2005). Many studies have found that stands with more residual trees have higher epiphyte abundance (Neitlich and McCune, 1997; Peck and McCune, 1997; Price et al., 1998; Peterson and McCune, 2001; Berryman and McCune, 2006). Although residual trees, particularly large veterans, had higher overall epiphyte abundance than regenerating trees in our second growth stands, we were unable to detect an effect of these remnant communities on epiphytic communities on younger trees for any functional group. We suggest two hypotheses for this inconsistency. First, assessment scale likely matters. Our study focused on tree-scale epiphytic communities supported by the dominant cohort, whereas we measured stand-level residual presence using randomly-located cruise plots that were not necessarily adjacent to felled trees. The presence of a residual tree, especially a larger veteran, close to the sampled tree would likely be a more sensitive indicator (Neitlich and McCune, 1997; Sillett and Goslin, 1999). Second, the quality of residual trees likely matters (Berryman and McCune, 2006). In harvested stands, trees were presumably left standing due to undesirable timber qualities rather than desirable ecological qualities; they may be too small, too few or too young to facilitate oldgrowth lichen community recovery. Benson and Coxson (2002) found lower lichen loading on smaller residual trees in interior rainforests. In our sample, even communities in stands with large veteran trees, however, did not differ from those with smaller or no residuals.

4.5. Management implications

Studies in other regions have suggested that long rotations, up to 200 years, might be sufficient to maintain epiphytic communities (Kuusinen and Siitonen, 1998; Price and Hochachka, 2001; Benson and Coxson, 2002). In the northern portion of the Great Bear Rainforest, however, epiphytic communities have not recovered to oldgrowth conditions by 200 years after harvest, remaining about 50% similar between 55 and 200 years. In common with other retrospective studies in coastal BC (e.g., Price et al., 1998; Price and Hochachka, 2001) second growth stands in this study were small, surrounded by oldgrowth and contained remnant trees. Larger clearcuts, established between the 1970s and early 2000, are more typical in the operable portions of the Great Bear Rainforest. Epiphytic recovery in these more uniform stands, lacking nearby propagule sources, will likely be even slower than indicated by our study. These sites thus may not meet oldgrowth recovery targets recently established in the Great Bear Rainforest agreement (e.g., oldgrowth representation across 70% of each ecosystem by the year 2264, MFLNRO, 2016), although some sites could have high richness even in young stands. Climate change increases the uncertainty of recovery trajectories. The operational 250-year-old definition for BC's coastal oldgrowth—originally used due to the practical difficulty of aging older trees and stands—may need to be reconsidered if the intent of representation is to maintain all oldgrowth attributes.

In the Great Bear Rainforest, western redcedar is common in oldgrowth, but occurs rarely in second growth stands (Banner and LePage, 2008). Without active management, this tree species will form a very minor component of managed coastal stands. We found that western redcedar hosted a different epiphytic community than other tree species; hence active management may increase stand-level epiphytic diversity.

Stand-level retention at the level proposed for the Great Bear Rainforest (15%, MFLNRO, 2016) may not facilitate recovery of epiphyte communities. The presence of remnant trees from the pre-harvest stand (as logged in the early to mid-1900s) did not detectably affect epiphyte community recovery in our study. Berryman and McCune (2006) concluded that 15% retention maintained fairly high abundance of hair and other lichens, but very few cyanolichens. Selecting the largest and most desirable trees to retain may provide increased benefits for epiphytes, although our study suggests that these will not necessarily colonise regenerating trees. Retaining trees of all species present pre-harvest (including western redcedar) and preferentially retaining amabilis fir (which hosted the highest diversity) could also benefit epiphyte diversity. In the absence of additional management strategies to promote epiphyte diversity, fully developed epiphyte communities in the northern region of the Great Bear Rainforest will be largely restricted to the remaining oldgrowth ecosystems.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foreco.2017.02.023.

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